17 July 1956

MILITARY SPECIFICATION
MOUNTS, GUN; LAUNCHING SYSTEMS, ROCKET AND GUIDED MISSILE; PROJECTORS, GENERAL SPECIFICATION

1. SCOPE

1.1 Scope. This specification presents the general requirements for the design, development and manufacture of gun mounts, rocket and guided missile launching systems and projectors which are under the cognizance of the Bureau of Ordnance. The general requirements of this specification, when accompanied by the requirements of a specific gun mount or launching system specification, will establish those parameters of the design and manufacture of a particular gun mount, launching system or projector to insure the delivery of equipment which will provide the performance required with minimum maintenance and maximum safety. The equipment covered by this specification will herein be referred to as "mounts and launching systems".

1.2 Classification. This specification shall apply to all classes and all types of mounts and launching systems.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards and publications, of the issue in effect on the date of invitation for bids, form a part of this specification to the extent noted herein.

SPECIFICATIONS

FEDERAL

QQ-N-290	Nickel Plating (Electrodeposited)
QQ-C-320	Chromium Plating (Electrodeposited)
QQ-Z-325	Zinc Plating (Electrodeposited)
QQ-S-365	Silver Plating (Electrodeposited)
QQ-P-416	Plating, Cadmium (Electrodeposited)

MILITARY

MIL-C-490	Cleaning and Preparation of Ferrous and Zinc Coated Surfaces for Organic Protective Coatings
MIL-S-901	Shockproof Equipment, Class HI (High-Impact) Shipboard Application, Tests for

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MIL-C-915	Cable, Cord and Wire, Electrical (Shipboard Use)
MIL-E-917	Equipment, Electric Power, Basic Requirements for (Naval Shipboard Use)
MIL-M-1940	Motors, Alternating Current (Shipboard Use)
MIL-R-2033	Relays for Electrical Equipment (Naval Shipboard Use)
MIL-E-2036	Enclosures for Electric and Electronic Equipment (Naval Shipboard Use)
MIL-C-2194	Cables, Electric, (Reduced Diameter Types) (Shipboard Use)
MIL-C-2212	Controllers, Motor Starters, and Master Switches - Alternating Current, Naval Shipboard Use
MIL-S-2335	Synchros, 60 Cycles, 115 Volts
MIL-M-2511	Motors, Direct Current, (Shipboard Use)
MIL-S-4040	Solenoid, Electrical, General Specification for
MIL-L-4343	Lubricating Grease, Pneumatic System
MIL-F-5509	Fittings, Fluid Connection
MIL-H-5512	Hose Assemblies, Aircraft Hydraulic and Pneumatic, High Pressure
MIL-P-5514	Packings; Installation and Gland Design of Aircraft Hydraulic and Pneumatic (General Specification for)
MIL-P-5516	Packings and Gaskets; Hydraulic, Aircraft
MIL-A-5541	Chemical Films for Aluminum and Aluminum Alloys

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MIL-T-8504	Tubing, Steel, Corrosion-Resistant (18-8) Annealed, Aircraft Hydraulic System
MIL-P-8564	Pneumatic Components, Aeronautical, General Specification for
MIL-A-8625	Anodic Coatings for Aluminum and Aluminum Alloys
MIL-C-16232	Coatings, Phosphate, Heavy (Manganese and Zinc Type) and Phosphate Treating Solutions
MIL-W-16878	Wire, Electrical (Insulated, High Temperature)
MIL-S-16892	Synchros, 400 cycle
MIL-I-16910	Interference Measurement, Radio Methods and Limits: 14 Kilocycles to 1000 Megacycles
MIL-M-17060	Motors; Alternating Current, Integral HP (Shipboard Use)
MIL-F-17111	Fluid, Power Transmission
MIL-T-17113	Tests, Shock, Vibration and Inclination for Electronic Equipment
MIL-S-17245	Synchros, 60 Cycle (Standardized)
MIL-H-18766	Hydraulic Equipment, Ordnance Shipboard
MIL-F-18870	Fire Control, Optical and Associated Equipment Intended for Shipboard Use, General Specification for Design and Procurement of
MIL-C-19052	Cleaning and Protecting Interior Non-machined Surfaces of Castings and Weldments

NAVY DEPARTMENT

42B5	Bearings, Ball
42B10	Bearings, Roller

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STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-108	Definitions of, and Basic Requirements for, Enclosures for Electric and Electronic Equipment
MIL-STD-129	Marking of Shipments
MIL-STD-167	Mechanical Vibration of Shipboard Equipment
MIL-STD-210	Climate Extremes for Military Equipment
MIL-STD-248	Qualification Test for Welders

BUREAU OF ORDNANCE

NAVORD OSTD 50	Preparation of Lubricating Charts
NAVORD OSTD 51	Method of Marking of Metal Scales for Sights and Fire Control Instruments
NAVORD OSTD 52	Painting of Ordnance Material
NAVORD OSTD 60	Six and Ten Splined Holes and Shafts
NAVORD OSTD 63	Piping, Tubing, and Fittings for Hydraulic Equipment
NAVORD OSTD 65	Ordnance Alterations and Modifications - Preparation, Issue, and Accomplishment
NAVORD OSTD 78	Preparation of Ordnance Classification of Defects
NAVORD OSTD 599	Drafting Practice

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PUBLICATIONS

BUREAU OF ORDNANCE

NAVORD OD 3000	Lubrication of Ordnance Material
NAVORD OD 8999	Provisioning Requirements of Initial Spares
NAVORD OD 10500	Preparation of Factory Acceptance Tests
OP 1	Preparation of Ordnance Publications
OP 400	General Instructions for the Design, Manufacture, and Inspection of Naval Ordnance Equipment
OP 1810	Preservation, Packaging, and Carloading for Shipment of Guns, Mounts, and Missile Launchers
OP 2230	Workmanship and Design Practices for Electronic Equipment
NAVORD OS 6341	Miscellaneous General Ordnance Design Requirements

BUREAU OF SHIPS

NAVSHIPS 250-435	Description and Application of Navy Standard Stuffing Tubes and Box Connectors
General Specification for Building Vessels of the United States Navy	Section S1-10 Noise, Shock and Vibration. Appendix 4, Parts 1 and 2 - Specification for Riveting-Appendix 5, Parts 1 and 2 - Specification for Welding

MISCELLANEOUS

DSM M5	Military Outline of Form and Instructions for the Preparation of Specifications

BUREAU OF ORDNANCE

Dwg. 564101	Terminal Boards
Dwg. 8-Z-1141	Terminal Boards, Assembly and Details

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(Copies of specifications, standards, and publications required by contractors in connection with specific procurement functions may be obtained from the procuring agency or as directed by the contracting officer.)

Ordnance Publications. Copies of Bureau of Ordnance specifications, drawings, standards, and other referenced Bureau publications, except Ordnance Pamphlets, may be obtained upon application to the Superintendent, U. S. Naval Gun Factory, Washington 25, D. C. Ordnance Pamphlets may be obtained from the Bureau of Ordnance, Department of the Navy, Washington 25, D. C.

Navy Publications. Copies of Military Specifications, Navy Department Specifications and Navy Shipment Marking Handbook may be obtained upon application to the Bureau of Supplies and Accounts, Navy Department, Washington 25, D. C. When making inquiry concerning specifications or drawings, both the title and identifying symbol should be stipulated.

2.2 Other Publications. The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on the date of invitation for bids shall apply.

National Bureau of Standards

Handbook H-28	Screw-Thread Standards for Federal Services
1950 Supplement to Handbook H-28	

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D. C.)

American Standards Association

ASA B5.15-1950	Involute Splines, Side Bearing
ASA B5.26-1950	Involute Serrations
ASA B6.1-1932	Spur Gear Tooth Form
ASA B6.6-1946	Gear Tolerances and Inspection
ASA B6.7-1950	Involute Fine-Pitch System
ASA B6.8-1950	Fine Pitch Straight Bevel Gears
ASA B6.9-1950	Design for Fine-Pitch Worm Gearing
ASA B29.1-1950	Transmission Roller Chains and Sprocket Teeth

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American Standards Association, cont'd

ASA B29.2-1950 Inverted Tooth (Silent) Chains and Sprocket Teeth

(Copies of these standards may be obtained from the American Society of Mechanical Engineers, 29 West 39th Street, New York 18, New York.)

3. REQUIREMENTS

3.1 General Requirements

3.1.1 Conformance. Mounts, launching systems, their components and associated equipments shall be designed, manufactured, and inspected in accordance with the instructions contained in OP 400, this specification and the applicable detail specification. All parts shall be manufactured in accordance with dimensions, tolerances, finish, and similar data as specified on applicable drawings and in applicable specifications which have been approved by the Bureau of Ordnance.

3.1.2 Space, Weight, Assembly and Installation considerations shall be in accordance with the instructions contained in OP 400. In addition, the out of balance moments shall be kept to the minimum consistent with proper operation.

3.2 Environmental Requirements

3.2.1 Climatic Conditions. Mounts and launching systems shall be designed and constructed for operation under the climatic extremes as set forth for "Operation Ship World-Wide" in MIL-STD-210. When necessary to meet these requirements the equipment shall be designed to meet requirements of Condition II specified in NAVORD OS 6341. For components within the ship's structure the minimum temperature at which specified performance is attained shall be 32°F but the equipments shall start and operate satisfactorily, but not necessarily at specified rates and accuracies at 0°F. Maximum protection shall be provided to prevent damage due to temperature, humidity, dust, sand, rain, snow, salt water spray and atmosphere, and winds. This protection shall not interfere with proper operation of the equipment.

3.2.2 Moisture and Fungus Resistance. All material used shall be selected for moisture and fungus resistance, or be treated to provide this resistance as required by NAVORD OS 6341.

3.3 Service Conditions

3.3.1 Vibration. Shipboard mounts and launching systems shall be designed to be fully operable at the specified firing rate under the loads imposed by the vibration of the class of ship on which it is installed, acting in combination with any other loads that may be imposed on the installation. Rigidly mounted components and equipments

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shall be designed so that they will not be damaged by resonant vibration in the range of the frequencies caused by propeller blade excitation. Applicable design data for mounts and launching systems weighing up to 100,000 pounds are contained in the General Specification for Building Vessels of the United States Navy. See also MIL-STD-167.

3.3.2 Resistance to Blast Damage

3.3.2.1 Shocks. Mounts, launching systems and components shall be designed to withstand the applicable Class HI (High Impact) shocks as set forth in Specification MIL-S-901 and MIL-T-17113.

3.3.2.2 Blast. Mounts, launching systems and components subject to blast due to firing of adjacent installed armament shall be designed to withstand blast loads of 1000 lbs. per sq. ft. unless otherwise specified.

3.3.3 Night Operation. Mounts and launching systems shall be capable of normal functioning at the specified firing rate at night or during a blackout.

3.3.4 Roll, Pitch and Yaw. Shipboard mounts and launching systems shall be capable of satisfactory operation during and after accelerations due to the rolling, pitching, and yawing of the ship. Data concerning the magnitude and extent of these forces and motions will be furnished by the Bureau of Ordnance.

3.3.4.1 Sea Loads. The exposed portions of mounts and launching systems shall be protected in accordance with the requirements of NAVORD OS 6341.

3.4 Materials

3.4.1 Selection. All materials for use in the equipment and those specified for the design shall be of a suitable quality and type so as to insure meeting of the other requirements in this specification. Materials shall be selected and specified in accordance with OP 400.

3.4.2 Critical Materials. Consideration shall be given to economy and availability of critical materials in times of National emergency. Critical and strategic materials shall be used only when essential for satisfactory operation of the system. When a critical or strategic material is used where a non-critical material was previously used, or when large quantities are used, a report thereof shall be submitted to the Bureau of Ordnance stating the application, reason, and quantity per unit.

Note: The list of strategic and critical materials shall be requested from the Bureau of Ordnance.

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3.4.3 Substitution. After a material has been approved by the Bureau of Ordnance any substitution shall require approval in writing, prior to manufacture or use, from the Bureau of Ordnance. Approval of such substitution shall not constitute a deviation from the requirements of this specification.

3.4.4 Flammable Materials. Unless specifically approved, materials shall not be used which support combustion and that are not self-extinguishing. Flammable impregnating and potting compounds may be used only if they are enclosed in fire-resistant containers.

3.4.5 Toxic Materials. Unless specifically approved, materials shall not be used which might produce harmful toxic effects under the various conditions encountered in service.

3.4.6 Material Finish

3.4.6.1 Surface Roughness. Cast and forged finishes and the surface roughness of machined surfaces shall be specified in a manner providing most economical manufacture consistent with proper operation and good appearance.

3.4.6.2 Protective Coatings.

3.4.6.2.1 Painting. When required by the applicable drawing(s), paint shall be applied in accordance with Standard NAVORD OSTD 52. Iron and steel surfaces shall be prepared for painting in accordance with Specification MIL-C-490. Zinc or cadmium plated surfaces to be painted shall be treated in accordance with QQ-Z-325 Type III or QQ-P-416 Type III respectively. Aluminum alloy surfaces shall be anodized in accordance with MIL-A-8625 or treated in accordance with MIL-C-5541 before painting.

3.4.6.2.2 Interior Surfaces. Castings, forgings, and weldments which are to contain working parts such as shafts, bearings, gears, etc., and hydraulic fluids, where contamination would adversely affect the functioning of mechanisms, or where corrosion would affect the structural strength, shall be cleaned and protected in accordance with the requirements of Specification MIL-C-19052.

3.4.6.2.3 Phosphate Coatings. Phosphate coatings, where specified, shall conform to the requirements of Specification MIL-C-16232.

3.4.6.2.4 Plating, in general, is not required on exterior surfaces, but it is usually acceptable provided that there is no danger of the plating flaking and causing damage to the equipment. It is much better to use corrosion resistant material or paint as provided for herein. All plated lap joints, riveted, screwed, or bolted together shall be plated before joining together, but after all machining operations have been completed. When making tack or spot weldments, plating may be removed from the weld areas prior to welding when it is necessary to

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develop maximum strength at the weld. Subsequent to welding operations, the areas shall be repaired by hot metal spraying or by the application of a sealing material to prevent entrance of moisture. Plating of metal surfaces shall be in accordance with the requirements of the applicable specification for each type as follows:

a.	Nickel	QQ-N-290
b.	Chromium	QQ-C-320
c.	Zinc	QQ-Z-325 Type II Class 2
d.	Silver	QQ-S-365
e.	Cadmium	QQ-P-416 Type II Class A

3.4.6.2.4.1 Silver plating shall not be used where arcing or corona is likely to occur.

3.4.6.2.4.2 Bright platings shall not be used on exterior surfaces.

3.5 Mechanical Equipment

3.5.1 Bearings. The use of anti-friction bearings, wherever practicable under the conditions of loading, is preferred. Ball bearings shall conform to the requirements of Navy Department Specification 42B5 and roller bearings shall conform to the requirements of Navy Department Specification 42B10.

3.5.2 Coupling of Shafts. The choice of the type of connection and the design of the parts shall be such that a stable joint will be secured under all conditions of service. Shafting for transmission of accurate angular motion, large torque loads, and reversing loads should be in accordance with the requirements of Standard NAVORD OSTD 60. Involute splines are preferable, and when used, shall conform to the recommendations set forth in Standard ASA B5.15-1950. Involute serrations, when used, shall conform to the recommendations set forth in Standard ASA B5.26-1950. Consideration shall be given to tapered shafts with keys in designs involving high torque reversing loads.

3.5.3 Gear Drives. Spur gear tooth forms shall conform to the recommendations of Standard ASA B6.1-1932 or ASA B6.7-1950. 20 degree pressure angle tooth forms are preferred. Fine-pitch straight bevel gears shall conform to the recommendations of Standard ASA B6.8-1950. Fine pitch-worm gearing shall conform to the recommendations of Standard ASA B6.9-1950. Measurements of allowances and tolerances for gears shall be in accordance with the recommended practices of Standard ASA B6.6-1946.

3.5.4 Housings. Gear drive and other mechanical enclosures shall be watertight (3 foot head) in accordance with the requirements of Standard MIL-STD-108 and Specification MIL-E-2036. Means shall be provided, such as thermometer wells, for conveniently determining the temperature of the lubricant within each housing. Threaded fastenings

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used within gear drive housings shall be securely fastened in place, in order to prevent gear drive damage due to loose parts.

3.5.5 Drain Holes. In non-tight structures drain holes shall be cut and water courses provided to prevent the accumulation and retention of liquids. Watertight structures shall be provided with drain holes with plugs.

3.5.6 Access and Inspection Openings. Equipments and assemblies shall be designed for convenient operation, maintenance and test, when installed, particularly with respect to units containing synchros and adjustable electric, hydraulic and mechanical devices. Units containing gears shall permit visual inspection of all gear teeth. Covered openings may be provided for this purpose, which shall expose to view at least two teeth at a time, preferably at the meshing point.

3.5.7 Threads. Unless otherwise specified, all threads shall be in accordance with National Bureau of Standard Handbook H-28, and supplements thereto. The class and fit of threads shall be specified on the drawings as outlined in NAVORD OSTD 599. Thread materials shall be selected to prevent seizure, scratching, or galling.

3.5.8 Locking Devices. All fastenings shall be vibration resistant. The use of locking devices such as cotter pins, lock washers, lock nuts, or wired heads for all threaded parts is required. All thread-locking devices, such as lock washers, shall be fabricated of corrosion resistant metals or shall be suitably protected. The use of set screws or locking screws shall be avoided wherever practicable.

3.5.9 Welding. Welding procedures shall conform to the recommendations set forth in the General Specifications for Building Vessels of the United States Navy, Appendix 5, Parts 1 and 2, Specification for Welding. Welders shall be qualified in accordance with the requirements of Standard MIL-STD-248.

3.5.10 Riveting. Riveting procedures shall conform to the recommendations as set forth in the General Specification for Building Vessels of the United States Navy, Appendix 4, Specification for Riveting.

3.5.11 Lubrication. Means for lubrication of all moving parts shall be provided. All lubrication points shall be easily accessible and long-term lubrication shall be provided wherever practicable. Lubrication points which are not readily accessible shall be lubricated by a centralized or automatic system. Bushings pierced by holes for lubrication shall be positively fastened to prevent movement where such movement would cut off lubrication. Reference shall be made to NAVORD OD 3000 for specific lubricants for various applications. The number of different lubricants shall be kept to a minimum. Forced lubrication systems shall be provided with a pressure gage or a sight feed indicator.

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3.5.12 Manual Drives. Unless otherwise specified, manual drives shall be provided for both train and elevation, for servicing and maintenance only. They shall have a permanent shaft and removable drive handles. Permanently installed drive handles shall be provided for mounts and launching systems which do not contain train and elevation power drives, and for the manual drives of equipment associated with mounts and launching systems. The gear ratios shall be such that one man can move the loaded mount or launching system in either train or elevation.

3.5.13 Power-off Brakes. Unless otherwise specified in the detail specification, mounts and launching systems shall be equipped with brakes which shall be automatically applied upon power failure or any other cessation of power. These shall be arranged so as to be capable of being released during manual operation. (See 3.5.12).

3.6 Pneumatic Equipment

3.6.1 General. Pneumatic components shall conform to the requirements of Specification MIL-P-8564. Pneumatic packing design shall conform to the requirements of Specification MIL-P-5514. Pneumatic tubing shall conform to the requirements of Specification MIL-T-8504. Pneumatic hose shall conform to the requirements of Specification MIL-H-5512.

3.6.2 Diesel Action Prevention and Oil Trap. Pneumatic systems shall be provided with flame arrestors to prevent diesel action within the system. A trap shall be provided to prevent the accumulation of oil or water in the system.

3.6.3 Pneumatic System Lubrication. The lubricant for pneumatic systems shall conform to the requirements of Specification MIL-L-4343. No other lubricant or organic compound shall be used in the pneumatic system, or in any application where contamination of the pneumatic system is possible.

3.7 Hydraulic Equipment

3.7.1 General. Hydraulic equipment shall conform to the requirements of Specification MIL-H-18766 and be designed for operation with the hydraulic fluid specified in Specification MIL-F-17111. The use of flexible hydraulic lines must be approved by the Bureau of Ordnance. Piping, tubing and fittings for hydraulic equipment shall conform to the requirements of Specifications MIL-T-8504, MIL-F-5509 and Standard OSTD 63. Screw connections in accordance with Specification MIL-F-5509 are permitted for tubing sizes one inch O.D. or 3/4 ips or below, flange connections shall be required for larger sizes. Packings and gaskets shall conform to the requirements of Specifications MIL-P-5514, MIL-P-5516 and Standard OSTD 63. Micronic type filters shall be provided in all hydraulic systems unless otherwise approved. No cadmium plating shall be used in contact with hydraulic fluid.

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3.7.2 Prevention of Trapped Air. Hydraulic equipment shall be designed with vents and valves to prevent the accumulation of trapped air in any part of the system. Vent plugs or valves shall be provided in order that the system may be checked to make sure that it is full of oil before putting the system into operation. Accessible low point drainage shall be provided.

3.7.3 Noise and Vibration. Noise and vibration, such as caused by turbulence or cavitation in the hydraulic equipment or by unbalanced moving parts, shall be minimized by proper design. Noise and vibration transmitted from other equipment shall be minimized by proper selection and design of mountings, couplings, and similar parts.

3.8 Electrical Equipment

3.8.1 Electric Power Equipment. The electric power equipment shall conform to the requirements of Specifications MIL-E-917, MIL-M-1940, MIL-M-2511 and MIL-M-17060. This equipment shall operate satisfactorily under a 5 per cent variation in power supply frequency and a 10 per cent variation in power supply voltage, or not more than 10 per cent variation of the two combined. However, maximum performance of the installation need be attained only at normal voltage and frequency. Variation in voltage and frequency within the limits stipulated above shall result in no more than a proportional change in response for control functions.

3.8.2 Controllers, Motor Starters, and Master Switches. All controllers, motor starters and master switches associated with the functioning of the mounts or launchers shall conform to the requirements of Specifications MIL-E-917 and MIL-C-2212. All associated control circuits, including motor control circuits, shall operate normally on nominal voltages of 115 volts root-mean-square (rms) alternating current.

3.8.3 Electronic Equipment. All electronic equipment shall conform to the requirements of Specification MIL-F-18870. Where practicable, magnetic or transistor types of amplifiers and controls are preferred to vacuum tube devices for reliability and shock reasons.

3.8.4 Relays. Electric relays shall be in conformance with the requirements of Specification MIL-R-2033.

3.8.5 Cable and Wiring. Unless otherwise specified, external cables connecting separately mounted equipment will not be furnished by the contractor. When furnished with the equipment, external cables shall be in accordance with Specification MIL-C-915 or MIL-C-2194 and shall have the ends sealed in accordance with NAVSHIPS 250-435. All internal wiring shall be furnished by the contractor and shall conform to Specification MIL-W-16878 or MIL-C-915. Each terminal shall be marked for proper and convenient identification during installation and maintenance. Wire and terminal designations shall be submitted to the Bureau of Ordnance for approval. Wire and cables shall be installed to avoid

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contact with rough or irregular surfaces or sharp edges, under service conditions. Wires shall not be bent sharply where they enter insulation material. Where wires run through holes in metal (partitions, shields, and the like), they shall be protected by grommets.

3.8.6 Synchros. Unless otherwise specified, synchro units will be furnished by the Government. Applicable information on synchro units is contained in the following Specifications: 60 Cycle-115 Volt Synchros, MIL-S-2335; 60 Cycle Standardized Synchros, MIL-S-17245; and 400 Cycle Synchros, MIL-S-16892.

3.8.7 Solenoids. Solenoids shall conform to the requirements of Specification MIL-S-4040.

3.8.8 Radio Interference Suppression. Radio interference produced by the power and control equipment shall be restricted in accordance with the requirements of Specification MIL-I-16910.

3.8.9 Electrical Enclosures. Electrical enclosures for mounts or launching systems shall be spraytight if below decks or within protective structures; submersible (15 foot head) if exposed to the weather; all to conform to the requirements of Standard MIL-STD-108 and Specification MIL-E-2036. Each enclosure shall be provided with a diagram of the internal wiring attached to the inside of the enclosure as specified in Specification MIL-E-2036.

3.8.10 Electric Control Panels, Devices, and Connection Boxes shall conform with the requirements of MIL-E-917 and MIL-F-18870. The principal electric power and control components such as electronic amplifiers, control, selector and transfer switches, relays, indicating lights, controllers, and similar devices shall be combined and integrated into a minimum number of control panels to save space and weight and to facilitate installation, operation and maintenance. Panels and connection boxes shall be of steel construction in general. Suitable terminal boards and terminal tubes or cable clamps shall be provided for all cables to be connected to the panels and boxes. Each control panel, unit and similar control device shall be provided with a diagram of the internal wiring, printed thereon, or enclosed in fungus resistant plastic sheets attached to the inside of the enclosure. Navy type cable terminal tubes, shall be provided for all electric power and control equipments. Terminal tubes are to be plugged during shipment and storage of the equipment.

3.8.11 Accessibility of Electrical Components. Electrical connections, switches, lamps, synchros and similar parts shall be readily accessible for adjustment, replacement, inspection, repair, etc. All control synchros shall have a convenient, simple and readily accessible vernier means of zeroing and positive clamping.

3.8.12 Terminal Boards. Terminal boards of standard Navy Types shown

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on BuOrd Dwg. Nos. 564101 or 8-Z-1141 shall be provided, in general, for all external connections in control panels, units and similar control devices. Each terminal shall be suitably marked for proper and convenient identification during installation and maintenance. The terminal boards and internal wiring shall be arranged as far as practicable, so that all wires of each external cable may be connected to a single or immediately adjacent board.

3.9 Fire Control Equipment. Unless otherwise specified, all fire control equipment shall conform to the requirements of MIL-F-18870 and NAVORD OSTD 51.

3.10 De-icing Equipment. All mounts and launching systems shall be protected against the formation of ice in accordance with the requirements of NAVORD OS 6341.

3.11 Ease of Operation. The components of mounts or launching systems which require manipulation shall be capable of being operated by personnel wearing winter weight clothing. Movement of parts of these components shall have a minimum of backlash or lost motion, consistent with normal operation and performance, so that no special attention is required to avoid errors from this source. All dials and scales used in operation shall be so located with respect to the normal position of the operating personnel that they may be easily read. Hand setting cranks shall be of a design and size to facilitate their operation; they shall be located, as far as practicable, adjacent to their associated dials in order to prevent unintended operation in action. Selection of controls, such as local, remote, shall be effected with ease and certainty.

3.12 Rapidity of Operation. The design of mounts and launching systems shall be such that orders, signals and operations can be obeyed instantly when shifting from standby to operating conditions. It shall be possible to introduce and establish a new set-up, or change the existing set-up, in a minimum time consistent with the accelerations and velocities inherent in the equipments of the particular installation. The speed of synchronization and accuracy of operation with respect to static and dynamic orders will be specified in the particular mount or launching system specification in terms of time and permissible error.

3.13 Safety of Personnel and Equipment. The design and arrangement of mount and launching system equipment shall provide for mechanical or electrical interlocks to prevent injury to personnel or damage to the equipment during normal operation of the equipment. The controls, and controlled circuits, including electrical, hydraulic and pneumatic, shall be designed, where practicable, to provide for sequential operation, and arranged so that operation of any control or combination of controls, in other than normal sequence, shall not operate the equipment nor cause

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injury to personnel or damage to any component. Power failure shall not initiate equipment motion, create any dangerous condition, or cause damage to the equipment. Interlocks shall be designed to leave the equipment in the safest of the various possible alternate conditions in case of interlock or power failure.

3.14 Reversibility. Unless otherwise specified in the detail specification mounts and launching systems shall be fully reversible to permit projectiles or missiles to be returned from the mount or launching system to the magazine area in local control.

3.15 Standard Parts and Components. Standard parts and components shall be selected in accordance with the requirements of OP 400.

3.16 Interchangeability. The design shall be such that all like pieces and like sub-assemblies of subsequent assemblies will be interchangeable. When satisfactory function of the part requires that operation can be obtained only by fitting at assembly, or by the performance of other special operations such as lapping, running-in, etc., the drawing shall clearly define this requirement by appropriate note. These special operations shall be restricted to an absolute minimum and in any case subject to BuOrd approval prior to incorporation in the final design.

3.17 Drawings, Specifications and Publications. When required by the individual specification or contract, the following drawings and other documents shall be furnished the Bureau of Ordnance.

3.17.1 Drawings for Design Approval shall be prepared in accordance with OP 400, and shall be submitted to the Bureau of Ordnance prior to commencing manufacture. They may be layouts, sketches or other forms in accordance with the contractor's engineering practice provided that all definitive and necessary information is clearly set forth.

3.17.1.1 Outline and Arrangement Drawings shall contain sufficient information on the form, principal dimensions, locations, etc., for the individual equipments and for the complete launching system, to permit evaluation of the proposed arrangement for proper and convenient production, shipping, installation, operation, tests and maintenance.

3.17.1.2 Assembly and Associated Detail Drawings shall be provided for individual equipments and components. Detail drawings for approval action need cover only those parts which are required for complete understanding of the particular design. However, complete information on materials including identification with reference to applicable specifications and/or commercial designation is required for all parts.

3.17.1.3 Diagrams shall include functional, gearing, hydraulic and

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electric diagrams and schematics with sufficient data, identification of elements and functional information to clearly delineate the action of all component parts of the equipments in the various phases of operation.

3.17.1.4 Calculations, Performance and Functional Data and similar information shall be submitted with design drawings for approval, and shall include stress and strength calculations, performance, calculations covering speed, torque, force, efficiency, temperature and hydraulic pressures, calculations covering electric power, weight, center of gravity and quantity of hydraulic fluid in the system and estimated leakage rates. Performance and functional data shall be prepared for each item having a specific functional output in relation to the overall performance of the system. This data shall be typed on letter size sheets satisfactory for reproduction by standard blueprint methods.

3.17.1.5 Deviations. Proposed deviations from the requirements of the applicable specifications shall conform to the requirements of OP 400. Any exceptions to the requirements of this specification regarding materials, workmanship, construction, arrangements, controls, operation, etc., shall be stated when submitting the drawings for approval. Unless the attention of the Bureau of Ordnance is directed to these exceptions, the approval of drawings will not constitute a waiver of any specification requirements. In no case shall the approval of drawings by the Bureau of Ordnance constitute a waiver of any specification requirement for performance of the material delivered.

3.17.2 Drawings for Record. Drawings for record in accordance with OP 400 shall be furnished to the Bureau of Ordnance with delivery of the first mount or launching system. Record drawings are to be prepared in accordance with NAVORD OSTD 599 and be of the types listed therein. The drawings for record required are "Master" and "End-Product" drawings as defined by NAVORD OSTD 599. Prior to delivery and approval of these drawings, they shall have been thoroughly checked for completeness and to eliminate interferences, errors, and overlapping tolerances. (Guides for use as check-off lists for product engineering and checking of drawings may be obtained from the Bureau of Ordnance upon request).

3.17.3 Lubrication Charts shall be prepared in accordance with NAVORD OSTD 50 and furnished to the Bureau of Ordnance with the first mount or launching system.

3.17.4 Specifications. If required by the contract, the contractor shall prepare the following specifications in accordance with Defense Supply Management Agency Manual DSM M 5.

3.17.4.1 Design Specifications. Design specification shall be prepared for the complete equipment and each unit which is separately

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mounted and which has a direct functional requirement relative to the performance of the system. These specifications shall be submitted for design approval with the drawings for design approval. They shall be complete detail specifications in all respects and contain all of the design requirements governing the specific details of the design based on this specification which will be reflected by the record drawings. Direct reference to applicable items in this specification may be made in these design specifications. These specifications may include at the contractor's discretion the data required under paragraph 3.17.1.4 above and the detailed test procedures necessary to determine satisfactory performance and acceptance. (Sample specifications of this type of specification for guidance may be requested from the Bureau of Ordnance.)

3.17.4.2 Procurement Specifications. These specifications shall be prepared for the same items specified in paragraph 3.17.4 above and delivered with the record drawings. They shall contain all of the requirements necessary for preproduction samples, manufacture, inspection, testing, preservation, packaging, packing and marking. (Sample format or specifications of this type may be requested from the Bureau of Ordnance.

3.17.5 Preliminary Ordnance Pamphlet (OP). If required by the contract, the contractor shall prepare and deliver with the first system, 25 preliminary instruction manuals for the operation, servicing, adjusting and maintenance of the complete launching system. These manuals shall include necessary safety precautions and other data necessary for the evaluation of the prototype equipment. Prior to delivery of the first prototype equipment, the contractor shall submit 7 proof copies to the Bureau of Ordnance for review and approval. It is recommended that these copies be doubled spaced, so that notations and corrections may be made as necessary.

3.17.6 Final Ordnance Pamphlet (OP). If required by the contract, the contractor shall prepare a final manuscript and art work and 9 copies thereof, of the final pamphlet and forward them to the Bureau of Ordnance for printing. The contractor shall maintain close liaison with the Bureau of Ordnance Publications Branch and cognizant technical branch in the preparation of this manuscript to assure completion and issue of this pamphlet with the first production units to be installed for service use. This publication must be complete in all respects as specified in OP 1 and instructions of the BuOrd Publications Branch.

3.17.7 Installation and Test Instructions. This information shall be prepared in reproducible form and in accordance with the instructions contained in OP 1. A preliminary draft, including explanations, reference to applicable installation drawings and safety precautions for the installation, adjustment and test of the mount and launching system

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shall be submitted to the Bureau of Ordnance for approval. Two copies shall be forwarded to the installing activity sufficiently in advance of delivery of the mount and launching system for expeditious activity planning and installation. Two copies of the approved instructions shall be forwarded with drawings for record. Copies also shall accompany the mount and launching system shipped for installation. These instructions are to be considered Government publications, and as such they may not be copyrighted.

3.17.8 NAVORD Ordnance Data. If required by the contract or by the nature of the mounts and launching systems, instructions as to installation, operation, etc. shall be prepared in the form of NAVORD OD's in accordance with the requirements of OP 1. These OD's shall be referenced on the drawings against which they apply by a note, and shall be delivered with the drawings for record. See 4.4.3.

3.17.9 Ordnance Classification of Defects. If required by the contract, the contractor shall prepare NAVORD OCD's in accordance with the instructions contained in STANDARD NAVORD OSTD-78. The NAVORD OCD's shall be delivered with the drawings for record.

3.17.10 Factory Acceptance Test Report. The contractor shall prepare a Factory Acceptance Test Report as an OD, in accordance with NAVORD OD 10500. This publication shall be furnished with the drawings for record for use in tests of production items.

3.17.11 Spare Parts List. If required by the contract, the contractor shall prepare a recommended preliminary spares list which may be required for maintenance replacement of parts. This spares list shall be prepared in accordance with NAVORD OD 8999 and submitted to the Bureau of Ordnance before delivery of the record drawings. The Contractor, Bureau of Ordnance and Ordnance Supply Office will prepare the final spares list after review of the preliminary list. The contractor shall not manufacture or deliver spare parts unless specified in the contract.

3.17.12 Ordnance Alterations - (NAVORD ORDALTS). After completion of the manufacture and shipping of the mounts and launching systems for installation, changes approved and considered necessary shall be made in accordance with instructions contained in NAVORD ORDALTS. These ORDALTS when required shall be prepared by the contractor in accordance with NAVORD OSTD 65 and forwarded to the Bureau of Ordnance.

3.18 Workmanship

3.18.1 General. The mount and launching system including all parts, components, accessories and equipment shall be constructed and finished in a manner which will assure meeting the requirements of this specification and the requirements of the individual specifications and drawings.

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Consideration and particular attention must be given to appearance, neatness, soldering, wiring, impregnation, plating, painting, riveting, assembly, welding, brazing and freedom of parts from burrs, sharp edges and corners. OP 2230 may be used as a guide for determining acceptable workmanship.

3.18.2 Gears. Gear assemblies shall be properly aligned and meshed, and shall be operable without interference, tight spots, loose spots or other irregularities. When required by the design for accurate adjustment, gear assemblies shall be free of backlash.

3.18.3 Clean-Up. All units and items shall be thoroughly cleaned and then lubricated if specified by the drawings. Loose, spattered or excess solder, metal chips, rosin flash and other foreign material or matter must be removed before final assembly and submission for Navy acceptance.

3.19 Name Plate Weight. The name plate weight is the aggregate of the scale weights of all the parts shown on the drawings which appear in the list of drawings of the component. Each name plate shall carry the name plate weight of the component to which it is affixed.

4. QUALITY ASSURANCE PROVISIONS AND TEST REQUIREMENTS

4.1 General. Satisfactory test data, indicating compliance with the specified requirements and suitability for the intended application, are required for acceptance of the mount and launching system and of any of its component equipments or units.

Mounts and launching systems and their individual components and equipment shall be inspected and tested in accordance with OP 400 and the outline of tests specified below. This inspection and testing shall be conducted at the contractor's plant, before shipment to a Naval activity and shall be performed in the presence of, and shall be approved by, a Naval Inspector or his representative. This inspection shall include a check for conformance with the requirements contained herein and for conformance with the design, dimensions, materials, finishes and treatments specified on the manufacturer's plans approved by the Bureau of Ordnance. The contractor shall supply all materials, equipment and personnel required for this purpose, and all tests shall be performed to demonstrate proper performance of the equipments to the satisfaction of the Inspector.

4.1.1 Unit Inspection. Navy inspection shall be applied to sub-assemblies and assembled units. Navy inspection of piece parts will be required only where such inspection is considered essential by the Bureau of Ordnance or the Inspector.

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4.1.2 Source Inspection of material procured under purchase orders and sub-contracts shall be limited to electric motors and such other major end products as may be designated by the Naval Inspector of Ordnance.

4.1.3 Failure to Meet Requirements. Failure of the mounts and launching systems, or of any of their components, to meet the specified requirements and tests shall require the contractor to effect the necessary adjustments, modifications, repairs or replacements, after which the material shall be submitted for further tests. Excessive wear, temperature rise, hydraulic leakage, noise and vibration are considered causes for rejection.

4.1.4 Sampling. Unless otherwise specified, and when applicable, the sampling plans and procedures used by the government inspector in the determination of the acceptability of products submitted by a supplier for government inspection, shall be in accordance with the provisions of MIL-STD-105.

4.2 Outline of Tests. The contractor shall prepare an outline of the tests to be conducted at his plant based on: the general requirements concerning the basic design, operating and functional characteristics and suitability of the individual hydraulic and electric power drive and control components; the detail requirements concerning the operational performance of complete equipments and systems, and of the launching system as a whole. The outline shall be submitted for approval, augmentation or modification by the Bureau of Ordnance as appears necessary in order to insure compliance with the requirements. Section 4 of Specifications MIL-F-18870, MIL-H-18766, and MIL-E-16400; the test requirements of Specifications MIL-M-1940, MIL-C-2212, and other specifications applicable to components; and the following shall be used as guides in the preparation of the outline of tests.

4.2.1 Tests Pertaining to General Requirements. All components shall be tested by the contractor to verify compliance with the general requirements specified herein, and to assure satisfactory characteristics and suitability of the individual units and components for their intended use in the launcher.

4.2.2 Tests Pertaining to Detail Requirements. The purpose of these tests is to check and verify the overall suitability and performance characteristics of the mount and launching system and its principal components, by means of measurements, observations, records and data covering time cycles, operating accuracies, noise and vibrations, endurance, wear, hydraulic temperature rise, buffer characteristics, electric power requirements, safety, convenience of control systems and equipments, etc. All rotating parts shall be checked to insure proper direction, speed,

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clearance and protection, and all actuating elements for interlocks, latches, and similar members shall be checked for proper operation. Lubrication, wear, noise and vibration conditions shall be observed, evaluated on the basis of acceptable performance and detailed in the test reports. Sampling plans and acceptance criteria for these tests will be provided by the procuring activity.

4.2.3 Tests at the Contractor's Plant. The tests at the contractor's plant shall be performed with the mounts and launching systems assembled and arranged on suitable stands to simulate, as far as practicable, the actual operating conditions for which they have been designed. Gearing, shafting and loading shall be such as to simulate, as far as practicable, the inertia, friction and natural frequencies of the driven systems as installed. All tests shall be performed with electrical power and control inputs at normal voltage and frequency. The quantity of fluid in the hydraulic equipments shall not exceed that intended for normal operations. Artificial means for cooling shall not be used unless specifically authorized by the Bureau of Ordnance or unless such means are provided for the actual equipments and intended for installation in the mount and launching system. Tests shall cover complete characteristics of the equipment including performance, power requirements, temperature rise, etc., under emergency conditions (power failure, emergency stop, etc.,) as well as normal conditions (start, warm-up, standby, operate, etc.).

4.2.4 Firing Tests. After satisfactory completion of the tests at the contractor's plant and prior to installation of the complete equipment, firing and operational tests may be conducted either at a Naval Proving Ground or at an equivalent firing range if this is found desirable during the course of development and production. The contractor shall assist in these tests with personnel and material as required. During firing, operational and feasibility tests, all mounts and launching systems shall be at the same adjustment used during the tests at the contractor's plant unless specifically approved otherwise by the Bureau of Ordnance. The service tests will be considered conclusive, and inferior performance, compared with the contractor's tests, may require reworking, readjustments and replacements, or may cause rejection of any unit or equipment involved.

4.3 Test Equipment

4.3.1 Contractor Furnished Equipment. The test equipment to be provided by the contractor for tests at his plant includes:

(1) The test stands mentioned above. Certain of the contractor's tests may require that the entire mount and launching system be assembled and operated on a

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stand which simulates as closely as practicable, the arrangement and mounting limitations of the type vessels on which installation of the equipment is contemplated. The design, fabrication and erection of the test stand structure shall require Bureau of Ordnance approval. The preparation of formal drawings covering such items will not be required.

(2) Accurate means for determining and recording:
Velocities of various components.
Angular position and accuracy in elevation and train.

(3) Accurate means for measuring and recording cyclic times as deemed necessary.

(4) Accurate means for measuring and recording pressures at all significant points.

4.3.1.1 Additional Equipment. If environmental (high and low ambient temperature, icing, etc.) test requirements exceed the capabilities of the contractor's plant, available facilities of a Naval or other activity shall be used instead of providing new test installations. The provision of equipment other than the contractor's plant facilities to provide the 1000 pound per square foot loading indicated herein for waves and blast loading is not contemplated.

4.3.2 Government Furnished Equipment

(1) Dummy missiles for cycling and operational checkout of the equipment will be furnished by the Government.

(2) Dummy directors for elevation and train control tests.

4.4 Reports. The tests performed at the manufacturer's plant shall be suitably recorded by means of data and graphs, and these records shall be included in the tests' reports.

4.4.1 Time Base. Test data which are to be recorded simultaneously by means of oscillograph, or similar recording means, shall have the same time base.

4.4.2 Identification. Zero time, calibrations, date, explanatory notes, test numbers, etc., shall be marked on all records for complete identification and understanding.

4.4.3 Form of Reports. The test reports shall be prepared in the

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form of Ordnance Data (NAVORD OD) (see 3.17.8) conforming to instructions contained in OP 1 and shall be reproducible. Copies of these reports shall be furnished as follows:

Bureau of Ordnance	2 copies
Naval Gun Factory	1 copy
Naval Inspector of Ordnance at Installing Activity	1 copy
Activity on which the launching system is to be installed	1 copy

5. PREPARATION FOR DELIVERY. The methods of preparation for delivery, packaging and packing of mounts and launching systems shall be subject to approval by the Bureau of Ordnance.

5.1 Packaging and Packing. Packaging and packing of mounts and launching systems shall be in accordance with Section 5 of Specification MIL-F-18870 and Specification MIL-H-18766 as applicable. All equipment shall be packaged and packed in such manner as to be acceptable to common carrier. All reasonable precautions shall be taken to prevent mechanical damage or damage due to dust or moisture, while the equipment is in transit. In preparation for shipment, all units shall be cleaned, lubricated, and coated with protective compounds, where necessary, and all openings in enclosures, such as inspection holes, terminal tubes, flange connections, and the like, shall be appropriately closed. Where deemed desirable, use may be made of standard Navy desiccant containers within fluid supply tanks of hydraulic equipments as a means of protecting internal components against moisture damage in shipment and pre-installation storage.

5.2 Marking. The exterior of each shipping container shall be marked in accordance with the applicable sections of MIL-STD-129.

5.3 Shipping. Acceptable practices, rules and equipment for shipment shall conform to the requirements of OP 1810.

6. NOTES

Patent Notice. When Government drawings, specifications or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever, and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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